

SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT: Artificial intelligence
PRACTICAL NO. 6

Aim: Adaboost Ensemble Learning

- Implement the Adaboost algorithm to create an ensemble of weak classifiers.
- Train the ensemble model on a given dataset and evaluate its performance.
- Compare the results with individual weak classifiers.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load the Iris dataset
df = pd.read_csv("Iris.csv")

# Select input features and target variable
X = df[['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']]
y = df['Species']

# Split the dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# Create a weak classifier using a decision stump
weak_model = DecisionTreeClassifier(
    max_depth=1,
```

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```
random_state=42
)

# Train the individual weak classifier
weak_model.fit(X_train, y_train)

# Predict using the weak classifier
weak_pred = weak_model.predict(X_test)

# Calculate weak classifier accuracy
weak_accuracy = accuracy_score(y_test, weak_pred)

# Create AdaBoost model
model = AdaBoostClassifier(
    estimator=weak_model,
    n_estimators=50,
    random_state=42
)

# Train the AdaBoost model
model.fit(X_train, y_train)

# Predict using AdaBoost
y_pred = model.predict(X_test)

# Calculate AdaBoost accuracy
adaboost_accuracy = accuracy_score(y_test, y_pred)

# Display accuracies
print("Weak Classifier Accuracy:")
print(weak_accuracy * 100, "%")
```

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```
print("\nAdaBoost Accuracy:")
print(adaboost_accuracy * 100, "%")

# Display classification report
print("\nAdaBoost Classification Report:")
print(classification_report(y_test, y_pred))

# Display confusion matrix
cm = confusion_matrix(y_test, y_pred)

print("\nAdaBoost Confusion Matrix:")
print(cm)

# Compare Weak Classifier and AdaBoost
models = ["Weak Classifier", "AdaBoost"]
accuracies = [weak_accuracy, adaboost_accuracy]

plt.figure(figsize=(8, 5))

plt.bar(models, accuracies)

plt.xlabel("Models")
plt.ylabel("Accuracy (%)")
plt.title("Weak Classifier vs AdaBoost")
plt.ylim(0, 1.05)

for i, value in enumerate(accuracies):
    plt.text(
        i,
        value + 0.01,
        f"{value:.2f}",
        ha="center"
    )

plt.tight_layout()
plt.show()
```

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```
# Visualize the AdaBoost confusion matrix
plt.figure(figsize=(6, 5))

plt.imshow(cm)

plt.title("AdaBoost Confusion Matrix")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.colorbar()

classes = model.classes_

plt.xticks(
    range(len(classes)),
    classes,
    rotation=20
)

plt.yticks(
    range(len(classes)),
    classes
)

# Display values inside the confusion matrix
for i in range(len(classes)):
    for j in range(len(classes)):
        plt.text(
            j,
            i,
            cm[i, j],
            ha="center",
            va="center"
        )

plt.tight_layout()
plt.show()
```

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OUTPUT:

```
IDLE Shell 3.13.2
File Edit Shell Debug Options Window Help
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: E:/AI/T071_AI_Prac.py =====
Weak Classifier Accuracy:
66.66666666666666 %

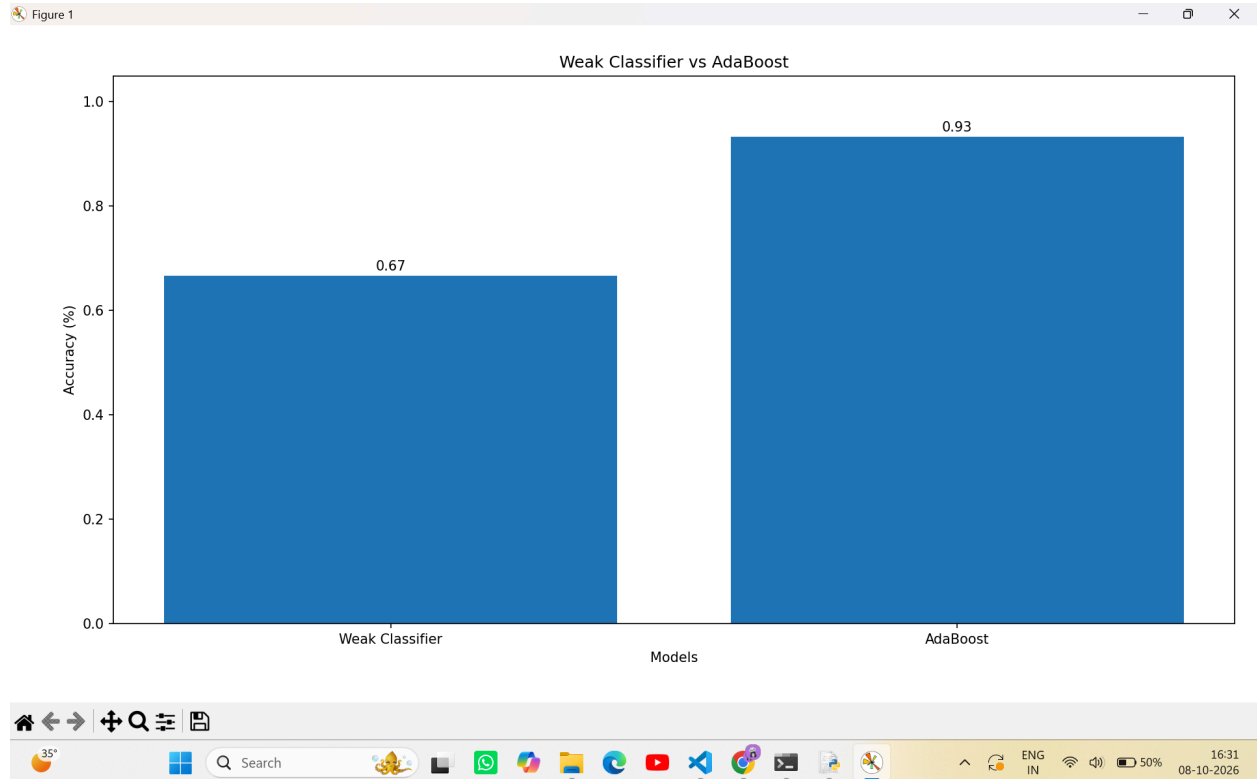
AdaBoost Accuracy:
93.33333333333333 %

AdaBoost Classification Report:
              precision    recall  f1-score   support

 Iris-setosa              1.00        1.00        1.00         10
 Iris-versicolor          0.90        0.90        0.90         10
 Iris-virginica           0.90        0.90        0.90         10

 accuracy                   0.93         0.93         0.93         30
 macro avg                  0.93         0.93         0.93         30
 weighted avg               0.93         0.93         0.93         30

AdaBoost Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  1  9]]
>>>
```



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